Fibonacci Periods Modulo Primes

Splitting Bounds and an Exact 2-Adic Restriction

Primary author (AI): ChatGPT (OpenAI)

Mathematical research note • 13 November 2026

Abstract

The OEIS sequence A060305 lists the periods of the Fibonacci sequence modulo successive primes. We give a self-contained finite-field proof of the classical bounds $\pi(p) \mid p-1$ or $\pi(p) \mid 2(p+1)$, depending on the residue of p modulo 5. In the nonsplit case we establish the stronger exact identity $v_2(\pi(p)) = v_2(p+1)+1$. All statements are known-type consequences of finite-field theory; no novelty is asserted.

Sequence and theorem

Let $F_0 = 0$, $F_1 = 1$, $F_{n+2} = F_{n+1} + F_n$. Define the *Pisano period* $\pi(m)$ as the least positive k such that

$$F_k \equiv 0 \pmod{m}, \qquad F_{k+1} \equiv 1 \pmod{m}.$$

The entry $a(n)$ of OEIS A060305 equals $\pi(p_n)$, where p_n is the n-th prime. The following divisibility restrictions are stated in the OEIS, with attribution to Wall [1, 2].

Theorem 1. *For every prime $p \notin \{2, 5\}$,*

$$\begin{aligned} p \equiv \pm 1 \pmod 5 \quad &\Longrightarrow \quad \pi(p) \mid p-1, \\ p \equiv \pm 2 \pmod 5 \quad &\Longrightarrow \quad \pi(p) \mid 2(p+1), \\ &\qquad\quad \pi(p) \nmid p+1, \quad v_2(\pi(p)) = v_2(p+1)+1. \end{aligned}$$

Here $v_2(n)$ denotes the exponent of 2 dividing the positive integer n. The exceptional values are $\pi(2) = 3$ and $\pi(5) = 20$.

Proof

Put

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_p), \qquad \det Q = -1.$$

Induction from the Fibonacci recurrence gives, for all $n \geq 1$,

$$Q^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \pmod p.$$

Since $F_{n-1} = F_{n+1} - F_n$, the relation $Q^n = I$ is *equivalent* to the two defining congruences for the Pisano period. Thus $\pi(p)$ is precisely the multiplicative order of Q.

Assume p is odd and $p \neq 5$. The characteristic polynomial $X^2 - X - 1$ of Q has discriminant $5 \neq 0$ in $\mathbb{F}_p$. Over $\mathbb{F}_{p^2}$ it has two distinct nonzero roots α, β with $\alpha\beta = -1$, so Q is diagonalizable with eigenvalues α, β.

If 5 is a square modulo p, both eigenvalues belong to $\mathbb{F}_p^\times$. Fermat's theorem yields $\alpha^{p-1} = \beta^{p-1} = 1$. Hence $Q^{p-1} = I$, and the order $\pi(p)$ divides $p-1$.

If 5 is a nonsquare modulo p, Frobenius interchanges the two roots: $\alpha^p = \beta$, $\beta^p = \alpha$. Consequently

$$\alpha^{p+1} = \alpha\beta = -1 = \beta^{p+1}, \qquad \text{and therefore} \qquad Q^{p+1} = -I.$$

As p is odd, $-I \neq I$, whereas $Q^{2(p+1)} = I$. Writing $s = p + 1 = 2^t u$ with u odd, the order $N = \pi(p)$ satisfies $N \mid 2s$ but $N \nmid s$. It follows directly that $v_2(N) = t + 1$.

Finally, quadratic reciprocity gives $(5/p) = (p/5)$ because $5 \equiv 1 \pmod 4$. Thus the square and nonsquare cases are respectively $p \equiv \pm 1$ and $p \equiv \pm 2$ modulo 5.

For $p = 2$, direct multiplication shows $Q^3 = I$ and $Q \neq I$, $Q^2 \neq I$, so $\pi(2) = 3$. For $p = 5$, write $Q = 3I + M$ over $\mathbb{F}_5$, where $M^2 = 0$ but $M \neq 0$. Then $Q = 3(I + 2M)$; the scalar factor has order 4 and the unipotent factor order 5, forcing $\pi(5) = 20$. □

Small-prime checks and scope

p	3	7	11	13	29
$\pi(p)$	8	16	10	28	14
$p \bmod 5$	3	2	1	3	4

The theorem gives divisibility restrictions, *not* an exact closed formula for each $\pi(p)$. In particular it does not settle Wall–Sun–Sun questions about periods modulo p^2. This is a self-contained exposition of classical results, with the 2-adic conclusion made explicit; no original discovery or independent human peer review is claimed.

References

[1] OEIS Foundation, *A060305: Pisano periods for primes*, https://oeis.org/A060305; see also https://oeis.org/A001175.

[2] D. D. Wall, "Fibonacci series modulo m," *American Mathematical Monthly* **67** (1960), 525–532.